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SCUOLA DI INGEGNERIA INDUSTRIALE
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Data Information Quality Project Report

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1 | Introduction

1.1. Presentation

The primary objective of this project is to implement a full data preparation and data analysis pipeline on the 'Comune di Milano - Servizi alla persona: parrucchieri ed estetisti' dataset. Data Analysis will be performed on both the dirty dataset and the cleaned dataset.

1.2. Methods and Techniques

The methodology follows a sequential approach composed of these steps:

1. **Data Profiling and Quality Assessment:** Analyzing the raw dataset to identify quality issues (missing values, outliers, inconsistencies).
2. **Data Cleaning:** Applying transformation rules to resolve identified issues.
3. **Data Analysis:** An analysis will be performed on both the original dirty dataset and the final cleaned dataset to compare results.

1.3. Setup and Libraries

The implementation leverages standard Python libraries for efficient data manipulation and visualization:

- **Pandas & NumPy:** Used for data structuring, ingestion, and numerical operations, providing the foundation for the data cleaning and transformation rules.
- **Matplotlib:** Employed for data visualization, specifically to plot distributions and null-value histograms during the profiling phase.
- **PyTorch:** Serves as the core framework for designing and training the hand-crafted Feed-Forward Neural Network used in the classification tasks.
- **Scikit-learn:** Utilized for data partitioning (train-test splits) and calculating performance metrics (Confusion Matrix, F1-Score, Precision, Recall).

2 | Pipeline Implementation

2.1. Data Profiling and Quality Assessment

A complete evaluation of the Data Quality dimensions of the dataset was carried out.

2.1.1. Completeness Evaluation

The number of null values in each column was extracted, and while the majority of the fields revealed a null rate lower than 1%, fields like 'Prevalente', 'Superficie Lavorativa' and 'Superficie Altri Usi' had a way higher null rate, with 'Prevalente' being the worst performing with a 92.5% null rate.

2.1.2. Accuracy Evaluation

Accuracy Evaluation was carried out for the fields 'ZD', that needs to be a number between 1 and 9, and 'Codice Via', a number between 100 and 9999. The dataset revealed a high level of accuracy, with both fields registering around 99.9% score.

2.1.3. Consistency evaluation

A consistency evaluation was carried out on 'Civico', 'Superficie Altri Usi' and 'Superficie Lavorativa', with each one of the fields scoring around 99.9%. Also, all the columns that share information with the 'Ubicazione' field were checked against it and revealed a 98% consistency rate.

2.2. Data Cleaning

2.2.1. Drop of 'Prevalente' and 'Tipo Esercizio PA'

The column 'Prevalente' is mainly composed of null values for the majority of its entries, therefore it was dropped and its information, combined with those of 'Tipo esercizio PA', were used to divide the elements into four categories. This was done by using a pattern that matched a set of keywords to the four macro categories, creating four Boolean columns with true/false values that better represented the information contained in 'Prevalente' and 'Tipo Esercizio PA'. Finally, the rows containing no true value in the new four columns were deleted, and the 'Prevalente' and 'Tipo Esercizio PA' were dropped.

2.2.2. Drop of 'Ubicazione'

The 'Ubicazione' column contained redundant information that was already present in 'Tipo Via', 'Via', 'Civico' and 'ZD'. In order to drop it, it was necessary to extract that information for rows that missed one or more of those fields.

By analyzing the dataset, it appeared that 77 rows had a missing value for one of those columns that could be extracted from 'Ubicazione' that could be retrieved using patterns and support columns.

After this process, the number of rows with missing values dropped from 77 to 1, and was therefore possible to drop the 'Ubicazione' column.

2.2.3. Duplicate Detection

In order to detect the duplicate entries, the dataset was analyzed on a combination of 'Tipo Via', 'Via' and 'Civico' to find activities with the exact same address in more than one entry.

By aggregating those rows, the dataset size dropped from 3909 to 3248 elements, meaning that 661 rows were duplicated and needed to be aggregated (17% of the total dataset).

2.2.4. Outliers Detection

We utilized the IQR method to detect and convert surface area outliers into NaN values.

This approach was chosen for its robustness against skewed distributions. It allowed us to replace erroneous entries with precise, category-based averages. We used an asymmetric version of IQR to not exclude legitimate values that are larger than usual, rather than a typing error.

The target columns were 'Superficie lavorativa' and 'Superficie altri usi', where respectively 59 and 25 outliers were found and substituted by NaN. Finally, missing data in 'Superficie altri usi' and 'Superficie lavorativa' were recovered by finding the mean of this parameter using all the available data from activities that belong to the same class (same Tipo Attività).

If the current row is "unique", the global mean of the column was provided.

3 | Results

3.1. Results Analysis

- **Completeness:** We achieved a significant improvement in completeness, rising from 79.03% to 100%. This was primarily driven by addressing the high null rates found in the "Prevalente" and "Superficie altri usi" variables, ensuring that no missing values remain in these fields by dropping the "Prevalente" field in favor of the four Boolean columns and statistically filling the missing values in "Superficie altri usi".
- **Accuracy:** The accuracy of the dataset was refined to perfection. Both the "ZD" and "Codice via" fields improved from high initial values (99.97% and 99.87% respectively) to 100%, eliminating all identified errors. This was achieved thanks to the extraction of the missing values from the "Ubicazione" column.
- **Consistency:** We also ensured high internal consistency. "Superficie altri usi" and "Superficie lavorativa" both reached 100% consistency (up from 99.87%). Additionally, inconsistencies in the "Civico" field were reduced, improving the metric from 99.92% to 99.94% of values expressed in the 000/A format.

Overall, the data cleaning strategies applied have resulted in a robust dataset that is now fully usable for analysis.

3.2. Data Analysis

This section analyzes the results of a ml data analysis performed on the dataset.

3.2.1. Prediction on dirty dataset

The primary objective of the experimental phase is to demonstrate the impact of data quality on model performance. To establish a baseline, the classification task is first addressed using the raw, "dirty" dataset. The goal is to classify businesses into specific service categories: 'Tipo A', 'Tipo B', 'Tipo C', 'Tipo D', based on information from other columns of the same activity. This is treated as a multi-label classification problem, as a single business can offer multiple services. The columns used to train the model are 'ZD', 'Tipo via', 'Superficie altri usi', and 'Superficie lavorativa', as they could provide useful information to the model for categorizing the different types of activities.

The model used for classification is a simple, handcrafted Feed Forward 2-Layer Neural Network, with adjustments made to avoid Overfitting, and to balance the classes' distributions, such as class weights. The performance on the uncleaned data serves as a reference point to evaluate the efficacy of the subsequent data preparation pipeline.

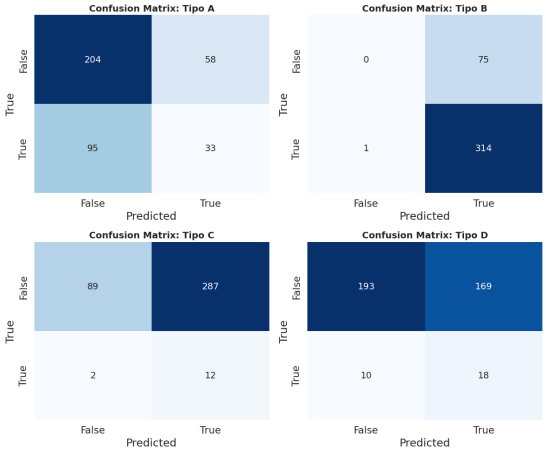


Figure 3.1: Dirty Dataset Confusion Matrix

- **Baseline Accuracy:** 11.54%
- **Baseline Precision:** 62.67%
- **Baseline Recall:** 77.73%
- **Baseline F1-Score:** 67.08%

Observations:
The matrix shows a strong bias toward the majority class (Tipo B) due to imbalance. Noise causes frequent misclassification, similar to random guessing, in minority classes (Tipo C/D), resulting in high recall but low accuracy due to overlapping definitions.

3.2.2. Prediction on cleaned dataset

The same classification model, with optimized hyperparameters, was subsequently trained on the cleaned dataset, following the Data Preparation Pipeline described in the previous chapters, with the addition of just a few operations made to adapt categorical columns and to normalize the dataset for performance enhancements.

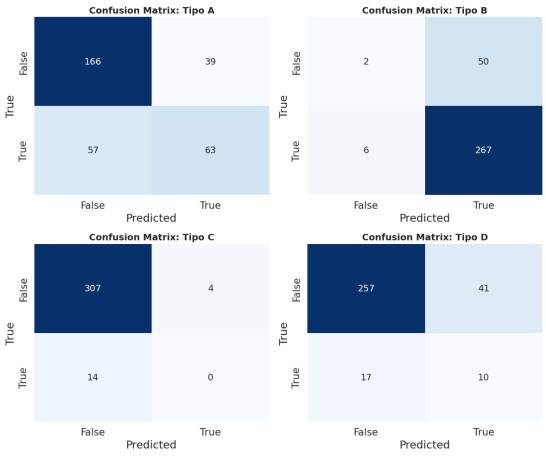


Figure 3.2: Clean Dataset Confusion Matrix

- **Accuracy:** 52.92
- **Precision:** 71.28%
- **Recall:** 78.34%
- **F1-Score:** 74.22%

Observations:
The matrix shows clearer class separation compared to the dirty baseline, with reduced noise in minority classes (Tipo C/D), and improved statistics. However, some confusion persists between overlapping categories, and the model still exhibits bias toward the dominant class (Tipo B), suggesting that class imbalance remains a challenge despite the cleaning efforts.

3.2.3. Challenges and Limitations

Despite the cleaning efforts, the classification results were limited by specific constraints:

- **Overfitting:** The available dataset size (<4000 rows) was insufficient to fully support the model's capacity (2 layers, 256 neurons). This disparity caused the model to memorize the training data rather than generalize effectively, triggering early stopping, to avoid overfitting.
- **Class Imbalance:** The dataset is heavily skewed towards 'Tipo B' (∼3132 samples) compared to rare classes like 'Tipo C' (∼140 samples). This imbalance creates a bias that is difficult to correct even with weighted loss functions.
- **Dataset specific problems:** The provided dataset proved inadequate for such an analysis. Only a small subset of columns were usable as features, and the total number of samples remained insufficient to achieve successful results.